

Integration of HYPR Learning Algorithm into non-spiking Recurrent Neural Networks (RNNs)

Daniel Siegfried Knapp
Felix Moitzi
Zlata Zelenska

TU Graz, IML
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Outline

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- HYPR gradient flow
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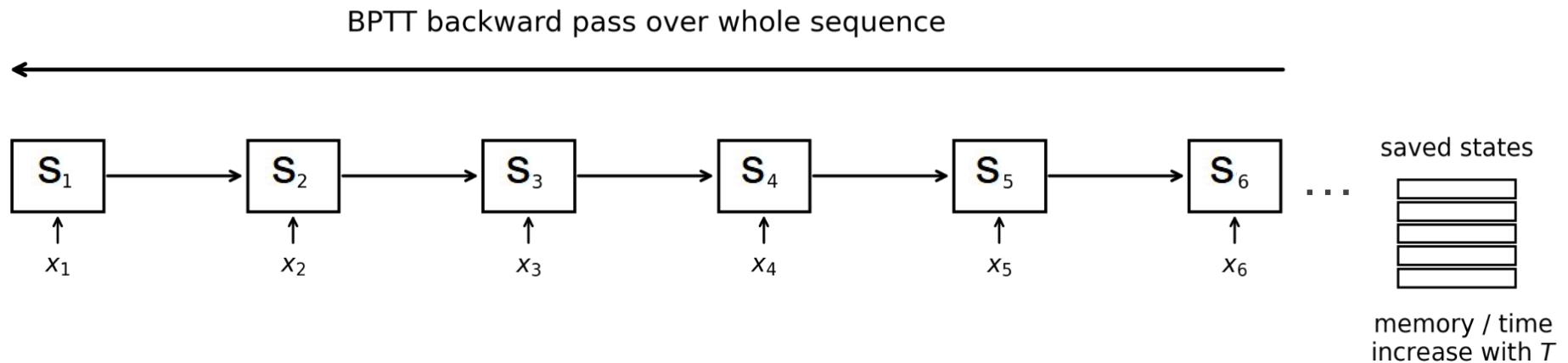
Problem definition

- RNNs are still strong models for sequential data
- **! But:** standard training uses **Backpropagation through time (BPTT)**

BPTT is **exact**, but also **sequential**, **offline**, and memory/time grow with sequence length

- **Research question:** can HYPR transfer from spiking RNNs to **non-spiking gated RNNs**?

Goal: implement HYPR for LSTM / GRU and compare it to BPTT



Background: Real-Time Recurrent Learning (RTRL) & E-prop

RTRL: exact online learning

- propagates total state sensitivity $P_t := \frac{ds_t}{d\theta}$
- avoids sequence-length-dependent memory

$$P_t = \frac{\partial f_\theta(s_{t-1}, x_t)}{\partial \theta} + \frac{\partial f_\theta(s_{t-1}, x_t)}{\partial s_{t-1}} P_{t-1}$$

$$\nabla_{\theta} \ell_t = \frac{\partial \ell_t}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial s_t} P_t$$

! But: P_t is large

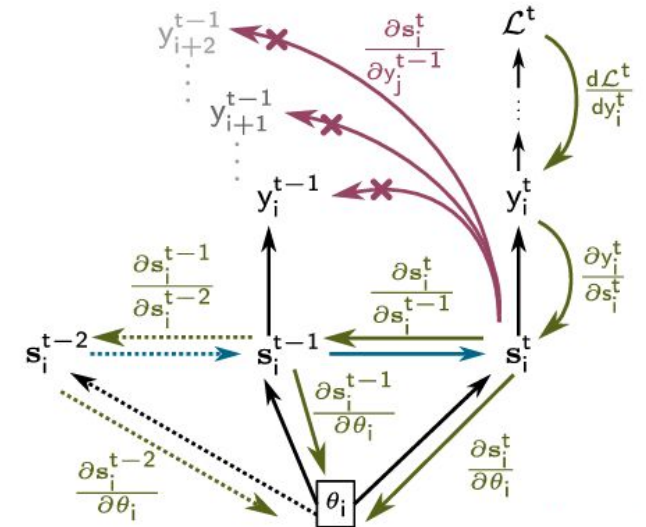
- cost scales with model size / parameter count
- exact online learning becomes impractical

E-prop: feasible approximation of RTRL

- keeps eligibility traces
- drops cross-neuron recurrent gradient paths
- originally proposed for spiking RNNs

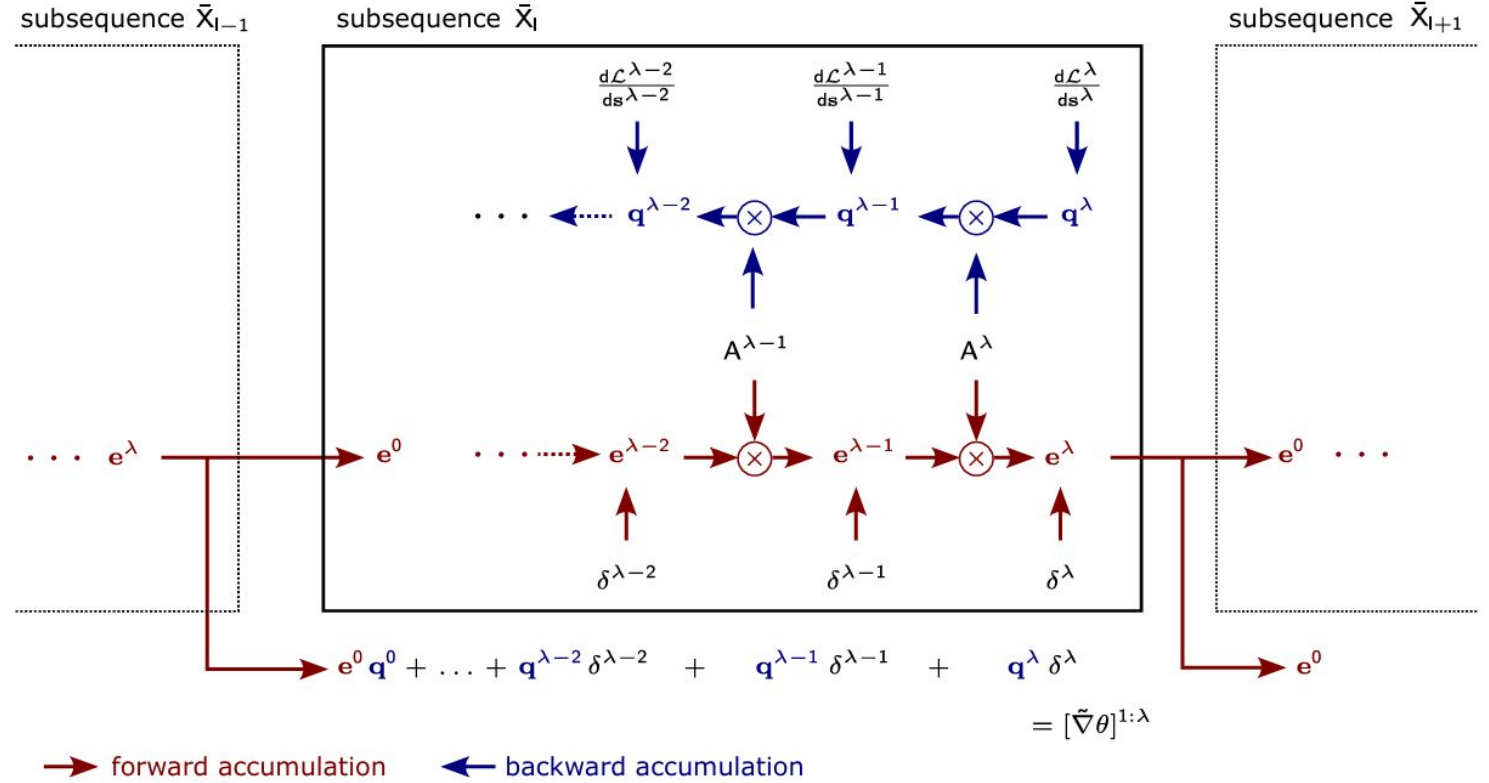
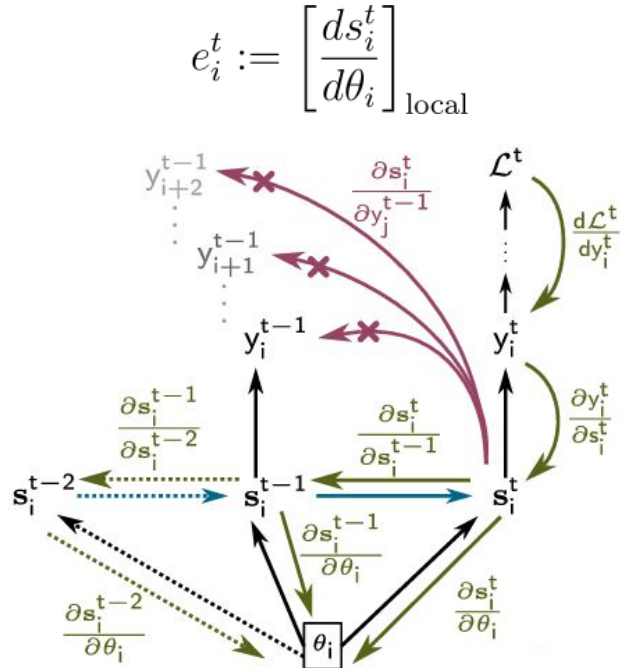
$$\tilde{\nabla}_{\theta_i}^t = \frac{d\mathcal{L}^t}{dy_i^t} \frac{\partial y_i^t}{\partial s_i^t} \left[\frac{ds_i^t}{d\theta_i} \right]_{\text{local}}$$

$$\left[\frac{ds_i^t}{d\theta_i} \right]_{\text{local}} \stackrel{\text{def}}{=} e_i^t = \frac{\partial s_i^t}{\partial s_i^{t-1}} e_i^{t-1} + \frac{\partial s_i^t}{\partial \theta_i}$$



HYPR gradient flow

* Schematic view: neuron index i is omitted in the graph for readability



$$A_i^t := \frac{\partial s_i^t}{\partial s_i^{t-1}}, \quad \delta_i^t := \left. \frac{\partial s_i^t}{\partial \theta_i} \right|_{\text{direct}}$$

$$e_i^t = A_i^t e_i^{t-1} + \delta_i^t$$

$$q_i^\lambda = \frac{\partial \ell_\lambda}{\partial s_i^\lambda}, \quad q_i^t = q_i^{t+1} A_i^{t+1} + \frac{\partial \ell_t}{\partial s_i^t}$$

$$[\tilde{\nabla}_{\theta_i}]^{1:\lambda} = q_i^0 e_i^0 + \sum_{t=1}^{\lambda} q_i^t \delta_i^t$$

HYPR algorithm

Split the full sequence into subsequences of length λ

S-stage (sequential):

- compute states, outputs, and losses inside the current subsequence

P-stage (parallel):

- compute outgoing eligibility e_θ^λ and pass it to the next subsequence
- calculate segment gradient $[\tilde{\nabla}\theta]^{1:\lambda}$
- Accumulate segment gradients, then update parameters

Algorithm 1: HYPR

Input: Time series X

Input: Network $\mathcal{N}$ with parameters θ

$e_\theta^0 \leftarrow 0$;

$\tilde{\nabla}\theta^{\text{cum}} \leftarrow 0$;

foreach Subsequence $\bar{X}_l$ in time series X **do**

$s^{1:\lambda}, y^{1:\lambda}, \mathcal{L}^{1:\lambda} \leftarrow \text{S-stage}(\mathcal{N}, \bar{X}_l)$;

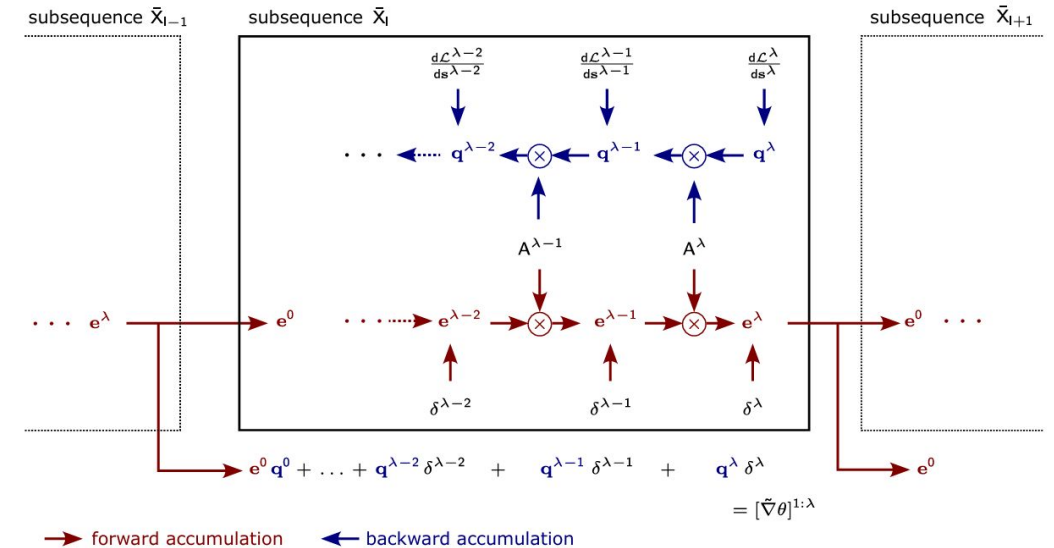
$e_\theta^\lambda, [\tilde{\nabla}\theta]^{1:\lambda} \leftarrow \text{P-stage}(\mathcal{N}, \bar{X}_l, s^{1:\lambda}, y^{1:\lambda}, e_\theta^0, \mathcal{L}^{1:\lambda})$;

$e_\theta^0 \leftarrow e_\theta^\lambda$;

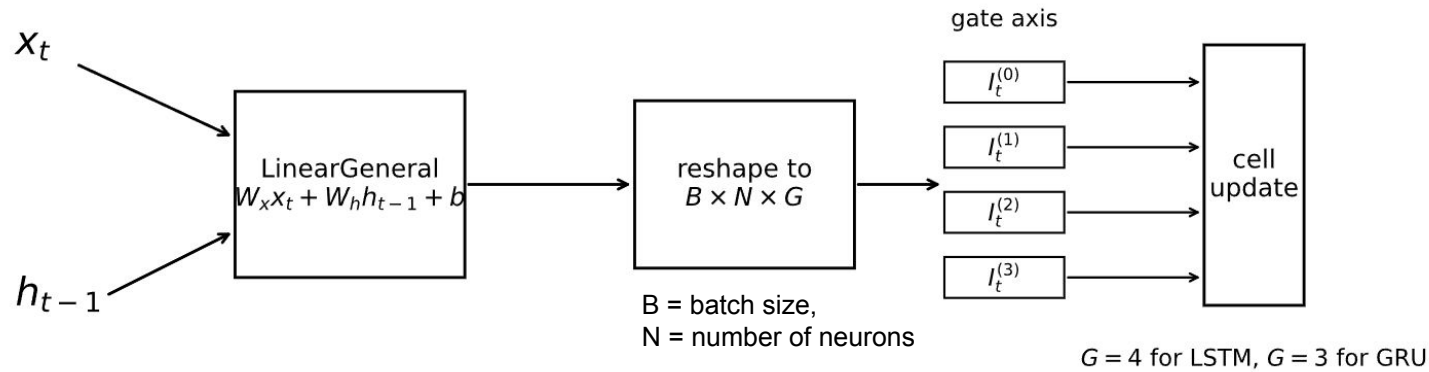
$\tilde{\nabla}\theta^{\text{cum}} \leftarrow \tilde{\nabla}\theta^{\text{cum}} + [\tilde{\nabla}\theta]^{1:\lambda}$;

end

$\theta \leftarrow \text{optimizer}(\tilde{\nabla}\theta^{\text{cum}})$;



Adapting HYPR to LSTM / GRU



$$s_t^{\text{GRU}} = h_t, \quad s_t^{\text{LSTM}} = (c_t, h_t)$$

$$z_t = W_x x_t + W_h h_{t-1} + b, \quad z_t \in \mathbb{R}^{B \times (NG)}$$

$$I_t = \text{reshape}(z_t, B, N, G) \in \mathbb{R}^{B \times N \times G}$$

- HYPR needs gate-specific Jacobians and eligibility tensors
- recurrent input projection is rewritten in a gate-aware form
- explicit gate axis:

$$I_t \in \mathbb{R}^{B \times N \times G}$$

Integration challenges

- **Shape stability**

With $\dim(I_t) = (B, NG)$ gate splitting using `.split()` can leave singleton dimensions $(B, N, 1)$. Direct indexing keeps gate tensors shape-stable

$$I_t \in \mathbb{R}^{B \times N \times G}, \quad \dim(I_t) = (B, N, G) \quad I_t^{(g)} = I_t[\dots, g] \in \mathbb{R}^{B \times N}$$

- **Eligibility-memory trade-off**

HYPR removes explicit dependence on total sequence length, but eligibility tensors scale with parameter count, state size, and number of gates

- **Chunk granularity / approximation trade-off**

More chunks increase parallel scope, but usually make the approximation harsher and can reduce accuracy

Grid search - Hyperparameter Configuration

- **Learning rate:** controls convergence speed and training stability
 - **Batch Size:** influences gradient quality and memory usage
 - **Hidden Size:** determines model capacity
 - **Number of Layers:** affects model depth
 - **Epochs:** how many epochs are trained (affects training time)
 - **HYPR: Number of Chunks:** determines partitioning of sequences in HYPR
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- Hyper-parameter configuration for tests is chosen by selecting run with best validation accuracy

Hyperparameter	Values
Learning Rate	{0.002, 0.001, 0.0005}
Batch Size	{64, 128, 256}
Hidden Size	{128, 256, 512}
Number of Layers	{1, 2, 3}
Epochs	{10, 20, 30}
Number of Chunks	{1, 2}

SHD & sMNIST

Hyperparameter	Values
Learning Rate	{0.002, 0.001, 0.0005}
Batch Size	{64, 128, 256}
Hidden Size	{128, 256, 512}
Number of Layers	{1, 2, 3}
Epochs	{50}
Number of Chunks	{1, 2}

Pathfinder-E

SHD & sMNIST datasets

SHD (Spiking Heidelberg digits):

- developed for benchmarking of SNN implementations
- 10.000 high-quality audio recordings of spoken digits (0-9)
- 50% male and 50% female speakers
- German and English Language

sMNIST:

- sequential MNIST dataset
- classification of handwritten Digits (0-9) from pixel sequences

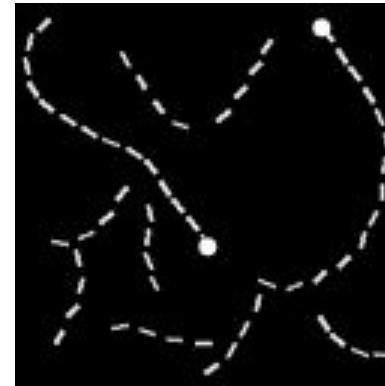


Pathfinder-E dataset

- generated by modifying “snakes2_wrapper.py” from the pathfinder repository
- images and metadata are compressed into difficult-level .pickle files and fed into the already existing dataset pipeline (used in BRF experiments)
- 32x32 and 128x128 resolution images
- classification task (are the points connected)
- test if model can capture information with long range dependencies / larger input data
- loss aggregation: per timestep cross entropy (last timestep cross entropy*)
- prediction mode: mean over sequence (last timestep prediction*)



32x32



128x128

sMNIST - LSTM

BPTT:

- validation acc: **0.9888** / test acc: **98.89%**
- best test acc: 98.94% (validation acc. 0.9864)

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.001	64	128	2	30

- test accuracy increases with increasing number of epochs
- more stable than HYPR (less overfitting on the validation set)

HYPR:

- validation acc: **0.9676** / test acc: **96.6%**
- this configuration also achieves the **max. test accuracy** in grid search!
- more sensitive to overfitting on the validation set
- with fixed hyper-parameter configuration no decrease in test accuracy visible
- validation acc: 0.9563 / test acc: 95.21% (28 chunks)
- (grid search max. test accuracy for 1 chunk: 95.93% with validation acc: 0.9624)

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.0005	128	512	2	30

Table 4.19: BPTT - Hyperparameter configuration LSTM

Number Epochs	10	20	30	50	100
Test Accuracy (%)	98.38	98.75	98.89	98.86	98.95
Validation Accuracy	0.9826	0.9868	0.9888	0.9898	0.9919

Table 4.20: BPTT's performance metrics across different epoch configurations

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.001	64	128	3	30	2

Table 4.22: HYPR - Hyperparameter configuration LSTM

Number Epochs	10	20	30	50	100
Test Accuracy (%)	93.73	96.37	96.60	97.00	96.93
Validation Accuracy	0.9433	0.9646	0.9676	0.9713	0.9739
Number of chunks	2	2	2	2	2

Table 4.23: HYPR's performance metrics across different epoch configurations

Number Chunks	1	2	4	8	16	28
Test Accuracy (%)	93.39	96.60	90.91	92.11	96.83	11.35
Validation Accuracy	0.9350	0.9676	0.9129	0.9271	0.9678	0.1115

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.002	64	128	2	30	28

Table 4.26: HYPR - Hyperparameter configuration LSTM - 28 Chunks

sMNIST - GRU

BPTT:

- validation acc: **0.9876** / test acc: **98.58%**
- best test acc: 98.79% (validation acc. 0.9864)

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.001	64	256	2	20

- slightly less validation / test accuracy than LSTM with BPTT

HYPR:

- validation acc: **0.9534** / test acc: **95.03%**
- this configuration also achieves the **max. test accuracy** in grid search! (for number-of-chunks = 1)
- HYPR needs proper hyper-parameter tuning !!
- with 2 chunks -> less validation / test accuracy

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.001	64	128	3	30	2

Number Epochs	10	20	30	50	100
Test Accuracy (%)	87.71	89.69	93.84	93.82	94.05
Validation Accuracy	0.8723	0.8923	0.9406	0.9458	0.9458
Number of chunks	2	2	2	2	2

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.002	128	128	3	30

Table 4.27: BPTT - Hyperparameter configuration GRU

Number Epochs	10	20	30	50	100
Test Accuracy (%)	98.24	98.79	98.58	97.70	97.70
Validation Accuracy	0.9796	0.9864	0.9876	0.9801	0.9801

Table 4.28: BPTT's performance metrics across different epoch configurations

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.001	64	256	3	30	1

Table 4.30: HYPR - Hyperparameter configuration GRU

Number Epochs	10	20	30	50	100
Test Accuracy (%)	90.15	94.62	95.03	95.03	95.03
Validation Accuracy	0.8995	0.9469	0.9534	0.9534	0.9534
Number of chunks	1	1	1	1	1

Table 4.31: HYPR's performance metrics across different epoch configurations

Number Chunks	1	2	4	8	16	28
Test Accuracy (%)	95.03	11.35	14.56	81.60	13.85	17.15
Validation Accuracy	0.9534	0.1115	0.1476	0.8193	0.1456	0.1706

Table 4.32: Performance metrics across different numbers of chunks

SHD - LSTM #1

BPTT

- (Best) validation acc 0.9963 / test acc 0.8838

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.002	256	256	3	30

Table 4.2: BPTT - Hyperparameter configuration LSTM

- validation acc 0.9791 / (best) test acc 0.9143

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.002	64	256	3	20

Table 4.4: BPTT - Hyperparameter configuration LSTM best test accuracy

- Best performance on different epoch-configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	86.3074	90.5035	88.3834	89.8852	87.9417
Validation Accuracy	0.9840	0.9950	0.9963	0.9914	0.9914

Table 4.3: BPTT's performance metrics across different epoch configurations

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HYPR

- (Best) validation acc 0.9914 / test acc 0.8401

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.001	64	512	3	30	2

Table 4.5: HYPR - Hyperparameter configuration LSTM

- validation acc 0.9914 / (best) test acc 0.8939

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.0005	64	512	3	30	2

Table 4.7: HYPR - Hyperparameter configuration LSTM best test accuracy

- Best performance on different epoch-configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	85.95	82.86	84.01	85.37	85.37
Validation Accuracy	0.9828	0.9901	0.9914	0.9901	0.9901
Number of chunks	2	2	2	2	2

Table 4.6: HYPR's performance metrics across different epoch configurations

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SHD - LSTM #2

HYPR

- Interest in number-of-chunks parameter
 - Other parameters same as in best run

Number Chunks	1	2	5	10	25
Test Accuracy (%)	55.03	84.01	83.78	81.66	83.65
Validation Accuracy	0.9877	0.9914	0.9889	0.9901	0.9877

Table 4.8: Performance metrics across different numbers of chunks

- Increasing of number-of-chunks not necessarily decrease in accuracy
- Parameter sensitive
 - e.g. best config. for 1 chunk → 0.8484 test accuracy
 - same parameters with 2 chunks → less accuracy

SHD - GRU #1

BPTT

- (Best) validation acc 0.9950 / test acc 0.8670

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.0005	64	512	3	30

Table 4.9: BPTT - Hyperparameter configuration GRU

- validation acc 0.9803 / (best) test acc 0.8962

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs
0.002	128	256	1	20

Table 4.11: BPTT - Hyperparameter configuration GRU best test accuracy

- Best performance on different epoch-configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	84.36	89.17	86.70	84.76	88.56
Validation Accuracy	0.9816	0.9938	0.9950	0.9938	0.9938

Table 4.10: BPTT's performance metrics across different epoch configurations

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HYPR

- (Best) validation acc 0.9889 / test acc 0.8568

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.0005	64	512	3	30	1

Table 4.12: HYPR - Hyperparameter configuration GRU

- validation acc 0.9718 / (best) test acc 0.8772

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.002	128	512	3	20	1

Table 4.14: HYPR - Hyperparameter configuration GRU best test accuracy

- Best performance on different epoch-configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	83.70	85.29	85.68	85.68	85.68
Validation Accuracy	0.9742	0.9840	0.9889	0.9889	0.9889
Number of chunks	1	1	1	1	1

Table 4.13: HYPR's performance metrics across different epoch configurations

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SHD - GRU #2

HYPR

- Interest in number-of-chunks parameter
 - Other parameters same as in best run

Number Chunks	1	2	5	10	25
Test Accuracy (%)	85.68	85.51	85.60	84.84	82.90
Validation Accuracy	0.9889	0.9852	0.9889	0.9828	0.9767

Table 4.15: Performance metrics across different numbers of chunks

- Increasing of number-of-chunks seems to decrease accuracy
- Parameter sensitive
 - e.g. best config. for 1 chunk → 0.8484 test accuracy
 - same parameters with 2 chunks → less accuracy

Comparison LSTM / GRU

- GRU best run with 2 number-of-chunks

LR	Batch Size	Hidden Size	Nr. of Layers	Epochs	Nr. of Chunks
0.001	64	512	2	30	2

Table 4.16: HYPR - Hyperparameter configuration GRU - direct comparison

Number Epochs	10	20	30	50	100
Test Accuracy (%)	83.70	85.29	84.36	84.36	84.36
Validation Accuracy	0.9742	0.9840	0.9852	0.9852	0.9852
Number of chunks	2	2	2	2	2

Table 4.17: HYPR's performance metrics across different epoch configurations

- Reminder LSTM:

Number Epochs	10	20	30	50	100
Test Accuracy (%)	85.95	82.86	84.01	85.37	85.37
Validation Accuracy	0.9828	0.9901	0.9914	0.9901	0.9901
Number of chunks	2	2	2	2	2

Table 4.6: HYPR's performance metrics across different epoch configurations

Pathfinder-E

LSTM:

- BPTT: validation acc: 0.5190 / test acc: 50.4%
- HYPR: validation acc: 0.5240 / test acc: 51.85%

GRU:

- BPTT: validation acc: 0.5190 / test acc: 50.4%
- HYPR: validation acc: 0.5240 / test acc: 51.85%
- accuracy metrics fluctuate around 50%
- constant high training loss
- Pathfinder -> binary classification task
- assumption: both models are not able to make meaningful predictions

Number Epochs	10	20	30	50	100
Test Accuracy (%)	50.05	50.70	49.15	50.40	49.15
Validation Accuracy	0.5135	0.5139	0.5200	0.5190	0.5200

Table 4.37: BPTT's performance metrics across different epoch configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	50.05	50.05	50.05	51.85	50.05
Validation Accuracy	0.5135	0.5135	0.5135	0.5240	0.5135
Number of chunks	2	2	2	2	2

Table 4.39: HYPR's performance metrics across different epoch configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	50.05	50.20	50.05	51.05	50.05
Validation Accuracy	0.5135	0.5135	0.5135	0.5210	0.5135

Table 4.41: BPTT's performance metrics across different epoch configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	50.65	50.20	50.05	50.05	47.70
Validation Accuracy	0.4895	0.4915	0.5305	0.5305	0.5220
Number of chunks	1	1	1	1	1

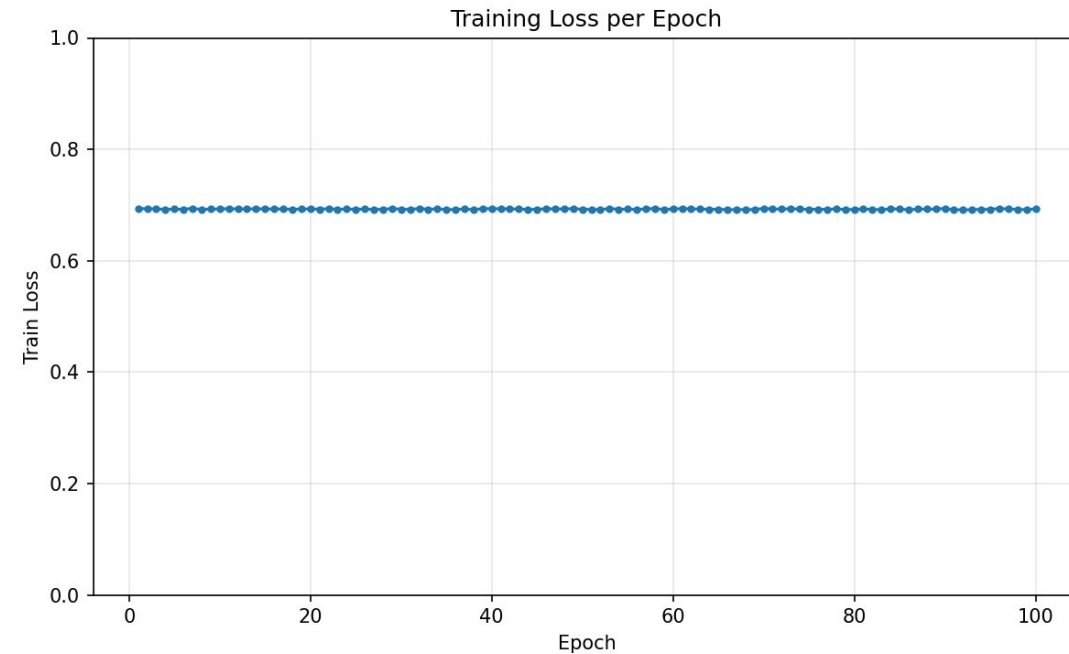
Table 4.43: HYPR's performance metrics across different epoch configurations

Number Epochs	10	20	30	50	100
Test Accuracy (%)	49.65	50.05	49.55	49.65	50.10
Validation Accuracy	0.5279	0.5135	0.5180	0.5279	0.5135
Number of chunks	2	2	2	2	2

Table 4.45: HYPR's performance metrics across different epoch configurations (comparison purposes)

Pathfinder-E

- generation of 128x128 dataset
- trying to overfit on smaller dataset (no success)

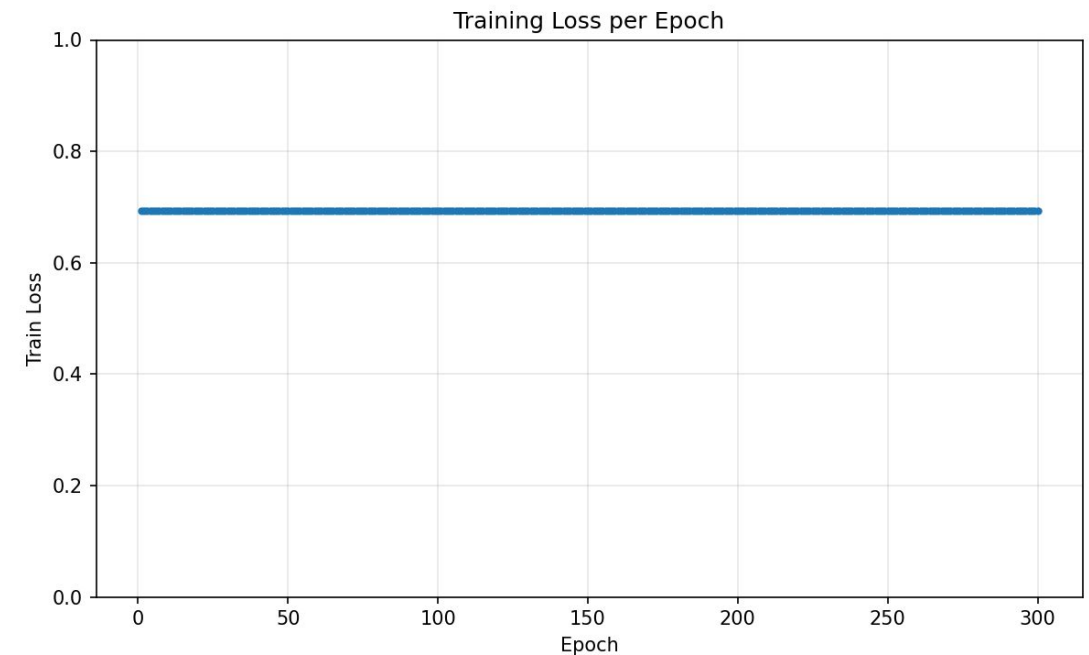


128x128 100 epochs

- constant training loss / no change in validation / test accuracy
- increase epochs to 300

Accuracy Metrics	BPTT	HYPR
Test Accuracy (%)	50.60	50.60
Validation Accuracy	0.5070	0.5070

Table 4.46: Pathfinder results on 128x128 input data (LSTM, 300 epochs)



128x128 300 epochs

Outcomes of the experiments

- **LSTM outperforms GRU** model with BPTT and HYPR
- **HYPR** can achieve **competitive performance** with both models
- **BPTT more stable** across different Hyper-parameter configurations
- Increasing the number of chunks -> decrease in performance (validation / test accuracy)
- proper **hyperparameter tuning** for **different number-of-chunks configurations** is important

Conclusion

- Can HYPR be transferred from RSNN to non-spiking gated RNNs: Yes
 - can be integrated into LSTM & GRU models
- But with gating more structure is needed → More computational power has to be handled
- Learning Dynamics:
 - No clear dominance → Tradeoff
 - BPTT stronger baseline → Gradient quality, robust optimization
 - HYPR stronger baseline → Bounded memory (respect to total sequence length), partial parallelization over subsequences
- “Winner” depending on real-world application setting